Performance Estimation Problems

A Tutorial – Session 1 / 3

March 2026

Aymeric Dieuleveut, Adrien Taylor

March 16, 2026

Outline

1. Introduction & Motivation
2. Reminders: Gradient Descent and Convergence Proof
3. Performance Estimation – Squared Distance
4. Performance Estimation – Function Value
5. The Generic PEP Framework & PEPit
6. Minimal Working Code in PEPit
7. Primal Feasible Points as Counter-Examples
8. Hands-on: Counter-Examples & Dimension Reduction

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→ **Take away**: for most first-order algorithms and simple function classes, we can get the tightest convergence rate numerically.

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Setting: first-order optimization

Minimize $f : \mathbb{R}^d \to \mathbb{R}$:

$$x_\star = \arg\min_{x \in \mathbb{R}^d} f(x),$$

where $f \in \mathcal{F}$ (encodes assumptions on the function class).

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Performance guarantees? Bounds on

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Regularity conditions: Smoothness

Definition : L-Smoothness

A differentiable function f is L-smooth if ∇f is L-Lipschitz:

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Proposition : Smooth + Convex characterization

A differentiable f is L-smooth and convex if and only if for all x, y:

$$f(y) + \langle \nabla f(y), x - y \rangle \;\leq\; f(x) \;\leq\; f(y) + \langle \nabla f(y), x - y \rangle + \tfrac{L}{2}\|x - y\|^2.$$

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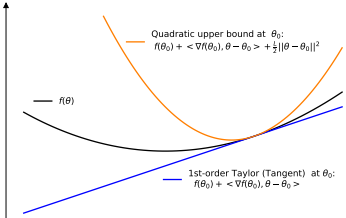

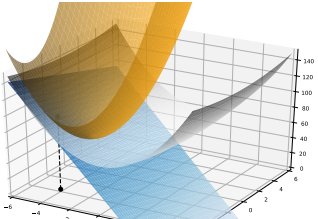

Observing f and ∇f *at* y provides information on the function *everywhere*.

Regularity conditions: Strong Convexity & the class $\mathcal{F}_{\mu,L}$

Definition : μ-Strong Convexity

f is μ-strongly convex if for all $x, y \in \mathbb{R}^d$:

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$\mathcal{F}_{\mu,L}$ denotes the set of μ-strongly convex and L-smooth functions on $\mathbb{R}^d$. (We allow $\mu = 0$ for convex-only.)

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Proposition : Co-coercivity (key inequality for GD analysis)

If $f \in \mathcal{F}_{\mu,L}$, then for all x, y:

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \;\geq\; \frac{\mu L}{\mu + L}\|x - y\|^2 + \frac{1}{\mu + L}\|\nabla f(x) - \nabla f(y)\|^2.$$

Gradient Descent: convergence rate

Theorem : Convergence of GD on $\mathcal{F}_{\mu,L}$

Let $f \in \mathcal{F}_{\mu,L}$ with $0 \leq \mu \leq L$.

GD with step size $\eta = \dfrac{2}{\mu + L}$ satisfies, after t iterations:

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→ The rate $\left(\dfrac{L-\mu}{L+\mu}\right)^2 \approx 1 - \dfrac{2\mu}{L}$ for $\kappa = L/\mu \gg 1$.

Proof of convergence (sketch)

Step 1. Expand the squared distance:

$$\|x_{t+1} - x_\star\|^2 = \|x_t - \eta \nabla f(x_t) - x_\star\|^2 = \|x_t - x_\star\|^2 - 2\eta \langle x_t - x_\star, \nabla f(x_t) \rangle + \eta^2 \|\nabla f(x_t)\|^2.$$

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Step 2. Apply co-coercivity (with $\nabla f(x_\star) = 0$):

$$\langle \nabla f(x_t),\, x_t - x_\star\rangle \;\geq\; \frac{\mu L}{\mu + L}\|x_t - x_\star\|^2 + \frac{1}{\mu + L}\|\nabla f(x_t)\|^2.$$

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$$\Rightarrow \text{By induction: } \|x_t - x_\star\|^2 \leq \left(\frac{L - \mu}{L + \mu}\right)^{2t}\|x_0 - x_\star\|^2.$$

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EXAMPLE ONE

—

Primal PEP for squared distance

Crash course on Performance Estimation Problems

$\rightarrow$ **Performance Estimation problems**: rethinking proofs of first-order optimization [1].

What is a worst-case guarantee (WCG)?

Example:

$$\forall f \in \mathcal{F}_{\mu,L}, \quad \text{for } x_1 = x_0 - L^{-1}\nabla f(x_0): \quad \|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2$$

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\end{aligned}$$

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Key insight The problem has been lifted to a semidefinite program (SDP) – a convex problem!

What did we just do? – Summary

$$\forall f \in \mathcal{F}_{\mu,L}, \quad \text{for } x_1 = x_0 - L^{-1}\nabla f(x_0): \quad \|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2$$

Step 1. Write the tightest τ as a supremum:

$$\tau^* = \sup_{\substack{f \in \mathcal{F}_{\mu,L},\ x_0, x_\star \\ x_1 = x_0 - L^{-1}\nabla f(x_0),\ x_\star = \arg\min f}} \frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2}$$

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Steps 2–3. Reduce to a finite dimensional problem (no more f):

- Sample $g_0 = \nabla f(x_0)$; set $x_\star = 0$ and $\|x_0\|^2 = 1$ by homogeneity.
- Replace $\exists f \in \mathcal{F}_{\mu,L}$ by **interpolation conditions** (Taylor et al. '17).

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Key point: the problem is now finite-dimensional (variables $x_0, x_1, g_0 \in \mathbb{R}^d$) and quadratic in (x_0, g_0).
$\rightsquigarrow$ One more step: lift to a convex SDP.

Step 4: lifting to a semidefinite program

Step 4. Every quadratic in (x_0, x_1, g_0) is **linear** in the Gram matrix $G = vv^\top$, $v = (x_0, x_1, g_0)$:

$$G = \begin{pmatrix} x_0 \\ x_1 \\ g_0 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ g_0 \end{pmatrix}^\top = \begin{pmatrix} \|x_0\|^2 & \langle x_0, x_1 \rangle & \langle x_0, g_0 \rangle \\ \langle x_1, x_0 \rangle & \|x_1\|^2 & \langle x_1, g_0 \rangle \\ \langle g_0, x_0 \rangle & \langle g_0, x_1 \rangle & \|g_0\|^2 \end{pmatrix} \succcurlyeq 0.$$

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The PEP becomes a standard SDP:

$$\tau^* = \sup_{\substack{G \succcurlyeq 0 \\ \mathrm{Tr}(G\, C_{\mathrm{alg}}) = 0 \\ G_{11} = 1 \\ \mathrm{Tr}(G\, C_{\mathrm{class}}) \geq 0}} G_{22}$$

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Basic take-away

→ **Convex SDP**: solvable with any solver.

Interactive Session #1

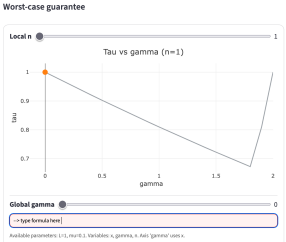
Exploring PEP numerically with the Streamlit UI

Activity: GD on L-smooth and μ-strongly convex functions — what is the worst-case rate for $\|x_1 - x_\star\|^2$?

1. Go to `pepitui-test.streamlit.app`
2. Select **Gradient Descent** (preimplemented), use *Customize* to set the performance metric to $\|x_1 - x_\star\|^2$ vs $\|x_0 - x_\star\|^2$
3. Set the function class to $\mathcal{F}_{\mu,L}$ (smooth **and** strongly convex)
4. Set $\mu = 0.1$, $L = 1$
5. Run `plot` and observe the worst-case value τ^*

$\Rightarrow$ **Try to guess the formula for** τ^* as a function of μ and L.

$\Rightarrow$ **Which step size gives the best rate?**



Outline

1. Introduction & Motivation
2. Reminders: Gradient Descent and Convergence Proof
3. Performance Estimation – Squared Distance
4. **Performance Estimation – Function Value**
5. The Generic PEP Framework & PEPit
6. Minimal Working Code in PEPit
7. Primal Feasible Points as Counter-Examples
8. Hands-on: Counter-Examples & Dimension Reduction

EXAMPLE TWO

—

Primal PEP for function-value criterion

PEP for $f(x_1) - f_\star \leq \tau \|x_0 - x_\star\|^2$

$$\forall f \in \mathcal{F}_{0,L}, \quad \text{for } x_1 = x_0 - \tfrac{1}{L}\nabla f(x_0): \quad f(x_1) - f_\star \;\leq\; \tau \|x_0 - x_\star\|^2$$

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&= \sup_{f,\, x_0, x_1, x_\star, g_0, g_1, f_0, f_1, f_\star} f_1 - f_\star \quad , \text{ s.t. } f \in \mathcal{F}_{0,L}, x_1 = x_0 - \tfrac{1}{L} g_0, \nabla f(x_\star) = 0 \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad g_0 = \nabla f(x_0),\ g_1 = \nabla f(x_1),\ f_i = f(x_i),\ f_\star = f(x_\star) \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \|x_0 - x_\star\|^2 = 1, \\
&= \sup_{x_0, x_1, x_\star, g_0, g_1, f_0, f_1} f_1 - f_\star \\
&\qquad \text{s.t.} \quad x_1 = x_0 - \tfrac{1}{L} g_0,\ \|x_0 - x_\star\|^2 = 1 \\
&\qquad\qquad \exists f \in \mathcal{F}_{0,L}:\ g_0 = \nabla f(x_0),\ g_1 = \nabla f(x_1), \\
&\qquad\qquad f_0 = f(x_0),\ f_1 = f(x_1),\ f_\star = f(x_\star),\ \nabla f(x_\star) = 0
\end{aligned}$$

PEP for $f(x_1) - f_\star$: interpolation and SDP lifting

Three points: x_0, $x_1 = x_0 - \frac{1}{L} g_0$, $x_\star$ ($g_\star = 0$). Interpolation on $\mathcal{F}_{0,L}$: $\forall\, i, j$:

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \frac{1}{2L} \|g_i - g_j\|^2.$$

PEP for $f(x_1) - f_\star$: interpolation and SDP lifting

Three points: $x_0,\ \ x_1 = x_0 - \frac{1}{L} g_0,\ \ x_\star\ (g_\star = 0)$. Interpolation on $\mathcal{F}_{0,L}$: $\forall\, i, j$:

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \tfrac{1}{2L} \|g_i - g_j\|^2.$$

SDP lifting. Gram matrix $G \succcurlyeq 0$ and function-value vector F:

$$G = \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix} \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix}^\top \succcurlyeq 0,\ \ G \in \mathbb{R}^{4\times 4}, \qquad F = \begin{pmatrix} f_0 \\ f_1 \\ f_\star \end{pmatrix} \in \mathbb{R}^3.$$

Each condition: $\mathrm{Tr}(G\, M_{ij}) + F^\top v_{ij} \geq 0$.

PEP for $f(x_1) - f_\star$: interpolation and SDP lifting

Three points: $x_0,\ x_1 = x_0 - \frac{1}{L} g_0,\ x_\star\ (g_\star = 0)$. Interpolation on $\mathcal{F}_{0,L}$: $\forall\, i, j$:

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \tfrac{1}{2L} \|g_i - g_j\|^2.$$

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Each condition: $\mathrm{Tr}(G\, M_{ij}) + F^\top v_{ij} \geq 0$.

$$\tau^* = \sup_{\substack{0 \preccurlyeq G,\ F \in \mathbb{R}^3 \\ \mathrm{Tr}(G\, M_{\mathrm{init}}) = 1 \\ \mathrm{Tr}(G\, C_{\mathrm{alg}}) = 0 \\ \mathrm{Tr}(G\, M_{ij}) + F^\top v_{ij} \geq 0}} F^\top v_{\mathrm{perf}}$$

PEP for $f(x_1) - f_\star$: interpolation and SDP lifting

Three points: x_0, $x_1 = x_0 - \frac{1}{L} g_0$, $x_\star$ ($g_\star = 0$). Interpolation on $\mathcal{F}_{0,L}$: $\forall\, i, j$:

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \tfrac{1}{2L} \|g_i - g_j\|^2.$$

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(i,j)	Condition
$(\star, 0)$	$f_\star \geq f_0 + \langle g_0, x_\star - x_0 \rangle + \frac{\|g_0\|^2}{2L}$
$(0, \star)$	$f_0 \geq f_\star + \frac{\|g_0\|^2}{2L}$
$(\star, 1)$	$f_\star \geq f_1 + \langle g_1, x_\star - x_1 \rangle + \frac{\|g_1\|^2}{2L}$
$(1, \star)$	$f_1 \geq f_\star + \frac{\|g_1\|^2}{2L}$
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$(1, 0)$	$f_1 \geq f_0 + \langle g_0, x_1 - x_0 \rangle + \frac{\|g_1 - g_0\|^2}{2L}$

$(\star, 0)$: $\mathrm{Tr}(G\, M_{\star 0}) + F^\top v_{\star 0} \geq 0$

$$M_{\star 0} = \begin{pmatrix} 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2L} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$v_{\star 0} = (-1, 0, +1)^\top$$

PEP for $f(x_1) - f_\star$: interpolation and SDP lifting

Three points: x_0, $x_1 = x_0 - \frac{1}{L} g_0$, $x_\star$ ($g_\star = 0$). Interpolation on $\mathcal{F}_{0,L}$: $\forall\, i, j$:

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \tfrac{1}{2L} \|g_i - g_j\|^2.$$

SDP lifting. Gram matrix $G \succcurlyeq 0$ and function-value vector F:

$$G = \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix} \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix}^\top \succcurlyeq 0, \quad G \in \mathbb{R}^{4\times 4}, \qquad F = \begin{pmatrix} f_0 \\ f_1 \\ f_\star \end{pmatrix} \in \mathbb{R}^3.$$

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$(\star, 0)$	$f_\star \geq f_0 + \langle g_0, x_\star - x_0 \rangle + \frac{\|g_0\|^2}{2L}$
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$(1, 0)$	$f_1 \geq f_0 + \langle g_0, x_1 - x_0 \rangle + \frac{\|g_1 - g_0\|^2}{2L}$

$(0, \star)$: $\mathrm{Tr}(G\, M_{0\star}) + F^\top v_{0\star} \geq 0$

$$M_{0\star} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2L} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$v_{0\star} = (+1, 0, -1)^\top$$

PEP for $f(x_1) - f_\star$: interpolation and SDP lifting

Three points: x_0, $x_1 = x_0 - \frac{1}{L} g_0$, $x_\star$ ($g_\star = 0$). Interpolation on $\mathcal{F}_{0,L}$: $\forall\, i, j$:

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \tfrac{1}{2L} \|g_i - g_j\|^2.$$

SDP lifting. Gram matrix $G \succcurlyeq 0$ and function-value vector F:

$$G = \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix} \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix}^\top \succcurlyeq 0, \quad G \in \mathbb{R}^{4\times 4}, \qquad F = \begin{pmatrix} f_0 \\ f_1 \\ f_\star \end{pmatrix} \in \mathbb{R}^3.$$

Each condition: $\mathrm{Tr}(G\, M_{ij}) + F^\top v_{ij} \geq 0$.

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(i,j)	Condition
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$(1, 0)$	$f_1 \geq f_0 + \langle g_0, x_1 - x_0 \rangle + \frac{\|g_1 - g_0\|^2}{2L}$

$(\star, 1)$: $\mathrm{Tr}(G\, M_{\star 1}) + F^\top v_{\star 1} \geq 0$

$$M_{\star 1} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & \frac{1}{2L} \end{pmatrix}$$

$$v_{\star 1} = (0, -1, +1)^\top$$

PEP for $f(x_1) - f_\star$: interpolation and SDP lifting

Three points: x_0, $x_1 = x_0 - \frac{1}{L} g_0$, $x_\star$ ($g_\star = 0$). Interpolation on $\mathcal{F}_{0,L}$: $\forall\, i, j$:

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \tfrac{1}{2L} \|g_i - g_j\|^2.$$

SDP lifting. Gram matrix $G \succcurlyeq 0$ and function-value vector F:

$$G = \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix} \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix}^\top \succcurlyeq 0, \quad G \in \mathbb{R}^{4\times 4}, \qquad F = \begin{pmatrix} f_0 \\ f_1 \\ f_\star \end{pmatrix} \in \mathbb{R}^3.$$

Each condition: $\mathrm{Tr}(G\, M_{ij}) + F^\top v_{ij} \geq 0$.

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(i,j)	Condition
$(\star, 0)$	$f_\star \geq f_0 + \langle g_0, x_\star - x_0 \rangle + \frac{\|g_0\|^2}{2L}$
$(0, \star)$	$f_0 \geq f_\star + \frac{\|g_0\|^2}{2L}$
$(\star, 1)$	$f_\star \geq f_1 + \langle g_1, x_\star - x_1 \rangle + \frac{\|g_1\|^2}{2L}$
$(1, \star)$	$f_1 \geq f_\star + \frac{\|g_1\|^2}{2L}$
$(0, 1)$	$f_0 \geq f_1 + \langle g_1, x_0 - x_1 \rangle + \frac{\|g_0 - g_1\|^2}{2L}$
$(1, 0)$	$f_1 \geq f_0 + \langle g_0, x_1 - x_0 \rangle + \frac{\|g_1 - g_0\|^2}{2L}$

$(1, \star)$: $\mathrm{Tr}(G\, M_{1\star}) + F^\top v_{1\star} \geq 0$

$$M_{1\star} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2L} \end{pmatrix}$$

$$v_{1\star} = (0, +1, -1)^\top$$

PEP for $f(x_1) - f_\star$: interpolation and SDP lifting

Three points: x_0, $x_1 = x_0 - \frac{1}{L} g_0$, $x_\star$ ($g_\star = 0$). Interpolation on $\mathcal{F}_{0,L}$: $\forall\, i, j$:

$$f_i \geq f_j + \langle g_j, x_i - x_j \rangle + \tfrac{1}{2L} \|g_i - g_j\|^2.$$

SDP lifting. Gram matrix $G \succcurlyeq 0$ and function-value vector F:

$$G = \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix} \begin{pmatrix} x_0 - x_\star \\ x_1 - x_\star \\ g_0 \\ g_1 \end{pmatrix}^\top \succcurlyeq 0, \quad G \in \mathbb{R}^{4\times 4}, \qquad F = \begin{pmatrix} f_0 \\ f_1 \\ f_\star \end{pmatrix} \in \mathbb{R}^3.$$

Each condition: $\mathrm{Tr}(G\, M_{ij}) + F^\top v_{ij} \geq 0$.

$$\tau^* = \sup_{\substack{0 \preccurlyeq G,\ F \in \mathbb{R}^3 \\ \mathrm{Tr}(G\, M_{\text{init}}) = 1 \\ \mathrm{Tr}(G\, C_{\text{alg}}) = 0 \\ \mathrm{Tr}(G\, M_{ij}) + F^\top v_{ij} \geq 0}} F^\top v_{\text{perf}}$$

(i,j)	Condition
$(\star, 0)$	$f_\star \geq f_0 + \langle g_0, x_\star - x_0 \rangle + \frac{\|g_0\|^2}{2L}$
$(0, \star)$	$f_0 \geq f_\star + \frac{\|g_0\|^2}{2L}$
$(\star, 1)$	$f_\star \geq f_1 + \langle g_1, x_\star - x_1 \rangle + \frac{\|g_1\|^2}{2L}$
$(1, \star)$	$f_1 \geq f_\star + \frac{\|g_1\|^2}{2L}$
$(0, 1)$	$f_0 \geq f_1 + \langle g_1, x_0 - x_1 \rangle + \frac{\|g_0 - g_1\|^2}{2L}$
$(1, 0)$	$f_1 \geq f_0 + \langle g_0, x_1 - x_0 \rangle + \frac{\|g_1 - g_0\|^2}{2L}$

$(0,1)$: $\mathrm{Tr}(G\, M_{01}) + F^\top v_{01} \geq 0$

$$M_{01} = \begin{pmatrix} 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & +\frac{1}{2} \\ 0 & 0 & -\frac{1}{2L} & +\frac{1}{2L} \\ -\frac{1}{2} & +\frac{1}{2} & +\frac{1}{2L} & -\frac{1}{2L} \end{pmatrix}$$

$$v_{01} = (+1, -1, 0)^\top$$

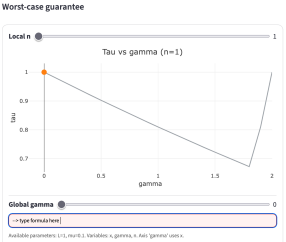
Interactive Session #2

Exploring PEP numerically with the Streamlit UI

Activity: GD on L-smooth convex functions — what is the worst-case rate for $f(x_1) - f_\star$?

1. Go to `pepitui-test.streamlit.app`
2. Select **Gradient Descent** (preimplemented) and function class $\mathcal{F}_{0,L}$ (smooth, $\mu = 0$)
3. Set $L = 1$, vary the step size γ
4. Performance metric: $f(x_1) - f_\star$ versus $\|x_0 - x_\star\|^2$
5. Run `plot` and observe the worst-case value τ^*

$\Rightarrow$ **Try to guess the formula for** τ^* as a function of L and γ.



Outline

1. Introduction & Motivation
2. Reminders: Gradient Descent and Convergence Proof
3. Performance Estimation – Squared Distance
4. Performance Estimation – Function Value
5. **The Generic PEP Framework & PEPit**
6. Minimal Working Code in PEPit
7. Primal Feasible Points as Counter-Examples
8. Hands-on: Counter-Examples & Dimension Reduction

The Generic PEP Framework

Example	$\forall f \in \mathcal{F}_{\mu,L}$	for $x_1 = x_0 - L^{-1}\nabla f(x_0)$	then	$\|x_1 - x_\star\|^2$	$\leq \tau \|x_0 - x_\star\|^2$
Generically	$\forall f \in \mathcal{F}$	for $x_T = \mathcal{A}(x_0, (\nabla f(x_t))_{t<T})$	then	$\mathrm{Perf}(x_T)$	$\leq \tau\, \mathrm{Init}(x_0)$
$\rightarrow$	Functional class	Algorithm		Worst-case guarantee	

The Generic PEP Framework

Example	$\forall f \in \mathcal{F}_{\mu,L}$	for $x_1 = x_0 - L^{-1}\nabla f(x_0)$	then	$\|x_1 - x_\star\|^2$	$\leq \tau \|x_0 - x_\star\|^2$
Generically	$\forall f \in \mathcal{F}$	for $x_T = \mathcal{A}(x_0, (\nabla f(x_t))_{t<T})$	then	$\mathrm{Perf}(x_T)$	$\leq \tau\, \mathrm{Init}(x_0)$
$\rightarrow$	Functional class	Algorithm		Worst-case guarantee	

Equivalently:

$$\tau^* = \max_{f,\, x_0,\, x_T} \frac{\mathrm{Perf}(x_T)}{\mathrm{Init}(x_0)}$$

s.t. $f \in \mathcal{F}$, $x_T = \mathcal{A}(x_0, (\nabla f(x_t))_{t<T})$

$\rightarrow$ Non-convex problem

$\rightarrow$ Infinite-dimensional over f

The Generic PEP Framework

Example	$\forall f \in \mathcal{F}_{\mu,L}$	for $x_1 = x_0 - L^{-1}\nabla f(x_0)$	then	$\|x_1 - x_\star\|^2$	$\leq \tau \|x_0 - x_\star\|^2$
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$\rightsquigarrow$ **Sampling, Interpolation, SDP lifting**

$$\tau^* = \sup_{\substack{0 \preccurlyeq G,\ F \in \mathbb{R}^p \\ \mathrm{Tr}(G\, M_{\mathrm{init}}) + F^\top v_{init} = 1 \\ \forall k,\ \mathrm{Tr}(G\, M_k) + F^\top v_k = 0}} \mathrm{Tr}(G\, M_{\mathrm{perf}}) + F^\top v_{\mathrm{perf}}$$

The Generic PEP Framework

Example	$\forall f \in \mathcal{F}_{\mu,L}$	for $x_1 = x_0 - L^{-1}\nabla f(x_0)$	then	$\|x_1 - x_\star\|^2$	$\leq \tau \|x_0 - x_\star\|^2$
Generically	$\forall f \in \mathcal{F}$	for $x_T = \mathcal{A}(x_0, (\nabla f(x_t))_{t<T})$	then	$\mathrm{Perf}(x_T)$	$\leq \tau\, \mathrm{Init}(x_0)$
$\rightarrow$	Functional class	Algorithm		Worst-case guarantee	

Equivalently:

$$\tau^* = \max_{f,\, x_0,\, x_T} \frac{\mathrm{Perf}(x_T)}{\mathrm{Init}(x_0)}$$

s.t. $f \in \mathcal{F}$, $x_T = \mathcal{A}(x_0, (\nabla f(x_t))_{t<T})$

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$$\tau^* = \sup_{\substack{0 \preccurlyeq G,\ F \in \mathbb{R}^p \\ \mathrm{Tr}(G\, M_{\mathrm{init}}) + F^\top v_{init} = 1 \\ \forall k,\ \mathrm{Tr}(G\, M_k) + F^\top v_k = 0}} \mathrm{Tr}(G\, M_{\mathrm{perf}}) + F^\top v_{\mathrm{perf}}$$

$\forall k$ aggregates algorithm & class (interpolation) constraints:

$$\begin{cases} \forall i,\ \mathrm{Tr}(G\, M_i^{\mathrm{alg}}) = 0 \\ \forall j,\ \mathrm{Tr}(G\, M_j^{\mathrm{class}}) + F^\top v_k \geq 0 \end{cases}$$

The Generic PEP Framework

Example	$\forall f \in \mathcal{F}_{\mu,L}$	for $x_1 = x_0 - L^{-1}\nabla f(x_0)$	then	$\|x_1 - x_\star\|^2$	$\leq \tau \|x_0 - x_\star\|^2$
Generically	$\forall f \in \mathcal{F}$	for $x_T = \mathcal{A}(x_0, (\nabla f(x_t))_{t<T})$	then	$\mathrm{Perf}(x_T)$	$\leq \tau\, \mathrm{Init}(x_0)$
$\rightarrow$	Functional class	Algorithm		Worst-case guarantee	

Equivalently:

$$\tau^* = \max_{f,\, x_0,\, x_T} \frac{\mathrm{Perf}(x_T)}{\mathrm{Init}(x_0)}$$

s.t. $f \in \mathcal{F}$, $x_T = \mathcal{A}(x_0, (\nabla f(x_t))_{t<T})$

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$$\tau^* = \sup_{\substack{0 \preccurlyeq G,\ F \in \mathbb{R}^p \\ \mathrm{Tr}(G\, M_{\mathrm{init}}) + F^\top v_{init} = 1 \\ \forall k,\ \mathrm{Tr}(G\, M_k) + F^\top v_k = 0}} \mathrm{Tr}(G\, M_{\mathrm{perf}}) + F^\top v_{\mathrm{perf}}$$

$\forall k$ aggregates algorithm & class (interpolation) constraints:

$$\begin{cases} \forall i,\ \mathrm{Tr}(G\, M_i^{\mathrm{alg}}) = 0 \\ \forall j,\ \mathrm{Tr}(G\, M_j^{\mathrm{class}}) + F^\top v_k \geq 0 \end{cases}$$

Worst-case guarantees with PEP

$\rightarrow$ Finding a tight WCG = solving a convex SDP

♡ Automatic numerical WCG $\rightarrow$ design & proof-checking

♡ Works for many algorithms & function classes

PEPit: computer-assisted worst-case analysis in Python

Goals of PEPit:

- Avoids manual SDP modeling steps
- Collaborative, easy-to-use methodology
- Code as close as possible to mathematical specification

Supported setups:

- Algorithms: Gradient, Prox, Inexact-Gradient, Line-search, ...
- Function classes: smooth, strongly convex, quadratically upper bounded, ...
- Performance metrics: $f(x_T) - f_\star$, $\|x_T - x_\star\|^2$, $\|\nabla f(x_T)\|^2$, ...

- **80+ documented examples**

- 1. Unconstrained convex minimization
  - 1.1. Gradient descent
  - 1.2. Subgradient method
  - 1.3. Subgradient method under restricted secant inequality and error bound
  - 1.4. Gradient descent with exact line search
  - 1.5. Conjugate gradient
  - 1.6. Heavy Ball momentum
  - 1.7. Accelerated gradient for convex objective
  - 1.8. Accelerated gradient for strongly convex objective
  - 1.9. Optimized gradient
  - 1.10. Optimized gradient for gradient
  - 1.11. Robust momentum
  - 1.12. Triple momentum
  - 1.13. Information theoretic exact method
  - 1.14. Proximal point
  - 1.15. Accelerated proximal point
  - 1.16. Inexact gradient descent
  - 1.17. Inexact gradient descent with exact line search
  - 1.18. Inexact accelerated gradient
  - 1.19. Epsilon-subgradient method
  - 1.20. Gradient descent for quadratically upper bounded convex objective
  - 1.21. Gradient descent with decreasing step sizes for quadratically upper bounded convex objective
  - 1.22. Conjugate gradient for quadratically upper bounded convex objective
  - 1.23. Heavy Ball momentum for quadratically upper bounded convex objective
- 2. Composite convex minimization
  - 2.1. Proximal gradient
  - 2.2. Accelerated proximal gradient
  - 2.3. Bregman proximal point
  - 2.4. Douglas Rachford splitting
  - 2.5. Douglas Rachford splitting contraction
  - 2.6. Accelerated Douglas Rachford splitting
  - 2.7. Frank Wolfe
  - 2.8. Improved interior method
  - 2.9. No Lips in function value
  - 2.10. No Lips in Bregman divergence
  - 2.11. Three operator splitting
- 3. Non-convex optimization
  - 3.1. Gradient Descent
  - 3.2. No Lips 1
  - 3.3. No Lips 2
- 4. Stochastic and randomized convex minimization
  - 4.1. Stochastic gradient descent
  - 4.2. Stochastic gradient descent in overparametrized setting
  - 4.3. SAGA
  - 4.4. Point SAGA
  - 4.5. Randomized coordinate descent for smooth strongly convex functions
  - 4.6. Randomized coordinate descent for smooth convex functions
- 5. Monotone inclusions and variational inequalities
  - 5.1. Proximal point
  - 5.2. Accelerated proximal point
  - 5.3. Optimal Strongly-monotone Proximal Point
  - 5.4. Douglas Rachford Splitting
  - 5.5. Three operator splitting
  - 5.6. Optimistic gradient
  - 5.7. Past extragradient
- 6. Fixed point
  - 6.1. Halpern iteration
  - 6.2. Optimal Contractive Halpern iteration
  - 6.3. Krasnosselskii-Mann with constant step-sizes
  - 6.4. Krasnosselskii-Mann with increasing step-sizes
- 7. Potential functions
  - 7.1. Gradient descent Lyapunov 1
  - 7.2. Gradient descent Lyapunov 2
  - 7.3. Accelerated gradient method
- 8. Inexact proximal methods
  - 8.1. Accelerated inexact forward backward
  - 8.2. Partially inexact Douglas Rachford splitting
  - 8.3. Relatively inexact proximal point
- 9. Adaptive methods
  - 9.1. Polyak steps in distance to optimum
  - 9.2. Polyak steps in function value
- 10. Low dimensional worst-cases scenarios
  - 10.1. Inexact gradient
  - 10.2. Non-convex gradient descent

- Complete docs, active community

What can we do with PEP?

Performance metrics: any linear combination of $f(x_T) - f_\star$, $\|x_T - x_\star\|^2$, $\|\nabla f(x_T)\|^2$ at various timesteps.

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For *all* of these, the worst-case rate is the solution of an SDP.

PEPit solves it in a few lines of Python.

EXAMPLES

—

Beyond gradient descent: more oracles in PEP

Example 1: Proximal Step to Minimize $f \in \mathcal{F}_{\mu,L}$

Algorithm. $x_{k+1} = \mathrm{prox}_{\gamma f}(x_k) := \arg\min_x \left\{ f(x) + \frac{1}{2\gamma}\|x - x_k\|^2 \right\}$.

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In PEP. Sample two triplets:

- $(x_k,\ g_k = \nabla f(x_k),\ f_k = f(x_k))$
- $(x_{k+1},\ g_{k+1} = \nabla f(x_{k+1}),\ f_{k+1} = f(x_{k+1})))$

Algorithm constraint: $x_{k+1} = x_k - \gamma\, g_{k+1}$

i.e. $\mathrm{Tr}(G\, C_{\mathrm{alg}}) = 0$ (linear in G).

Class constraint:

Interpolation conditions for $f \in \mathcal{F}_{\mu,L}$ on triplets $(x_k, g_k, g(x_k))$ and $(x_{k+1}, g_{k+1}, g(x_{k+1}))$.

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- `f = problem.declare_function(SmoothStronglyConvexFunction, mu=mu, L=L)`
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💡 - The implicit equation

$$x_{k+1} = x_k - \gamma \nabla f(x_{k+1})$$

is *not solved explicitly*: it is encoded as an algorithm constraint (+ interpolation conditions)

No closed form for $\mathrm{prox}_{\gamma f}$ is needed.

Example 2: Projection onto a Convex Set

Algorithm. Projected gradient descent:

$$x_{k+1} = \Pi_C(x_k - \gamma \nabla f(x_k)), \qquad \Pi_C(y) = \arg\min_{x \in C} \|x - y\|.$$

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Connection to PEP. The indicator function $\iota_C(x) = 0$ if $x \in C$, $+\infty$ otherwise, satisfies:

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Indicator functions of bounded sets, and of balls are interpolable

$\rightarrow$ see Convex indicator, Pepit doc.

Example 3: Inexact Gradient Step

Algorithm. Instead of the exact gradient, use an inexact oracle $\tilde{g}_k$ satisfying:

$$\|\tilde{g}_k - \nabla f(x_k)\| \leq \delta \quad \text{or} \quad \|\tilde{g}_k - \nabla f(x_k)\|^2 \leq \delta^2 \|\nabla f(x_k)\|^2.$$

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In PEP:

- Sample exact gradient $g_k = \nabla f(x_k)$ (interpolation conditions for f).
- Introduce $\tilde{g}_k$ as an **additional variable**.
- The inexactness condition is a **quadratic constraint** in G:

$$\|\tilde{g}_k - g_k\|^2 \leq \delta^2 \;\Leftrightarrow\; \mathrm{Tr}(G\, M_{\text{inexact}}) \leq \delta^2.$$

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💡 inexactness = one extra quadratic constraint.

PEP & PEPit handle this via `problem.add_constraint(...)`.

→ see Inexact gradient step oracle, Pepit doc.

Example 4: Line Search

Algorithm. Exact line search along the gradient direction:

$$\gamma_k = \arg\min_{\gamma \geq 0} \; f(x_k - \gamma \nabla f(x_k)), \qquad x_{k+1} = x_k - \gamma_k \nabla f(x_k).$$

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$$f(x_{k+1}) \;\leq\; f(x_k - \gamma\, g_k) \quad \forall\, \gamma \geq 0.$$

This is an **infinite** family of constraints: one for every γ.

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It's a relaxation!

Orthogonality is *necessary* for exact line search.

The feasible set of PEP is **strictly larger** than the set of true line-search instances.

$\rightarrow$ Rates are **valid** but not necessarily **tight**.

Other examples

See the documentation.

Primitive steps

- Inexact gradient step
- Exact line-search step
- Proximal step
- Inexact proximal step
- Bregman gradient step
- Bregman proximal step
- Linear optimization step
- Linearly shifted optimization step
- Epsilon-subgradient step

Figure: Primitive steps in Pepit

Outline

1. Introduction & Motivation
2. Reminders: Gradient Descent and Convergence Proof
3. Performance Estimation – Squared Distance
4. Performance Estimation – Function Value
5. The Generic PEP Framework & PEPit
6. **Minimal Working Code in PEPit**
7. Primal Feasible Points as Counter-Examples
8. Hands-on: Counter-Examples & Dimension Reduction

Minimal PEPit Example: GD on smooth convex f – MWE

Goal: find the tightest τ such that $f(x_n) - f_\star \leq \tau \|x_0 - x_\star\|^2$ for GD with step $\gamma = 1/L$.

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from PEPit import PEP
from PEPit.functions import SmoothConvexFunction

# --- Problem parameters ---
L, gamma, n = 1.0, 1.0, 3  # smoothness, step size, nb of steps

# --- Set up the PEP ---
problem = PEP()

# Declare the function class
func = problem.declare_function(SmoothConvexFunction, L=L)

# Define the optimum x_star and f_star
x_star = func.stationary_point()
f_star = func(x_star)

# Define the starting point
x0 = problem.set_initial_point()

# Initial condition: ||x0 - x_star||^2 <= 1
problem.set_initial_condition((x0 - x_star)**2 <= 1)

# Run n gradient steps
x = x0
for _ in range(n):
x = x - gamma * func.gradient(x)

# Set performance metric: f(x_n) - f_star
problem.set_performance_metric(func(x) - f_star)

# Solve the SDP
tau = problem.solve(verbose=1)
print(f"Worst-case rate: {tau:.6f}")  # should give L/(4n+2) =
    1/6 for n=1
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Figure: Pepit Output

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(PEPit) Final upper bound (dual): 0.0714281380988 9529 and lower bound (primal example): 0.0714275603704179
(PEPit) Duality gap: absolute: 5.697284773814593e-07 and relative: 7.976311586548534e-06
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Figure: Pepit Output

PEPit: what happens under the hood?

What you write (math):

$$\tau^* = \max_{f,\, x_0,\, x_T} \frac{\mathrm{Perf}(x_T)}{\mathrm{Init}(x_0)}$$

$$\text{s.t. } f \in \mathcal{F}, \;\; x_T = \mathcal{A}(x_0, (\nabla f(x_t))_{t<T})$$

What PEPit solves (SDP):

$$\tau^* = \sup_{0 \preccurlyeq G,\, F \in \mathbb{R}^p} \mathrm{Tr}(G\, M_{\mathrm{perf}}) + F^\top v_{\mathrm{perf}}$$

$$\mathrm{Tr}(G\, M_{\mathrm{init}}) + F^\top v_{init} = 1$$

$$\forall k,\; \mathrm{Tr}(G\, M_k) + F^\top v_k = 0$$

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PEPit automatically:

1. Builds the Gram matrix G from the declared points.
2. Encodes algorithm constraints as linear equalities in G, as well as other constraints.
3. Encodes interpolation conditions as linear inequalities in G.
4. Calls an SDP solver (e.g. MOSEK, SCS, CVXPY).
5. Returns the optimal rate + primal certificate (worst-case function).

PEPit: what happens under the hood?

What you write (math):

$$\tau^* = \max_{f,\, x_0,\, x_T} \frac{\mathrm{Perf}(x_T)}{\mathrm{Init}(x_0)}$$

$$\text{s.t. } f \in \mathcal{F}, \;\; x_T = \mathcal{A}(x_0, (\nabla f(x_t))_{t<T})$$

What PEPit solves (SDP):

$$\tau^* = \sup_{\substack{0 \preccurlyeq G,\, F \in \mathbb{R}^p \\ \mathrm{Tr}(G\, M_{\mathrm{init}}) + F^\top v_{init} = 1 \\ \forall k,\, \mathrm{Tr}(G\, M_k) + F^\top v_k = 0}} \mathrm{Tr}(G\, M_{\mathrm{perf}}) + F^\top v_{\mathrm{perf}}$$

PEPit automatically:

1. Builds the Gram matrix G from the declared points.
2. Encodes algorithm constraints as linear equalities in G, as well as other constraints.
3. Encodes interpolation conditions as linear inequalities in G.
4. Calls an SDP solver (e.g. MOSEK, SCS, CVXPY).
5. Returns the optimal rate + primal certificate (worst-case function).

Output for $n = 1$, $L = 1$, $\gamma = 1/L$

- Worst-case rate: $\tau^* \approx 0.1667$

Outline

1. Introduction & Motivation
2. Reminders: Gradient Descent and Convergence Proof
3. Performance Estimation – Squared Distance
4. Performance Estimation – Function Value
5. The Generic PEP Framework & PEPit
6. Minimal Working Code in PEPit
7. **Primal Feasible Points as Counter-Examples**
8. Hands-on: Counter-Examples & Dimension Reduction

What does the primal SDP tell us?

The PEP primal SDP (generic form):

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with the list corresponding to the list of algorithmic and class constraints: $\mathrm{Tr}(G\, C_{\ell,\text{alg}}) = 0, \mathrm{Tr}(G\, M_{ij}) + F^\top v_{ij} \geq 0,\ \forall i, j$

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$$G = \begin{pmatrix} x_0 \\ \vdots \\ g_k \end{pmatrix} \begin{pmatrix} x_0 \\ \vdots \\ g_k \end{pmatrix}^\top \succcurlyeq 0$$

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$\Rightarrow$ explicit algorithm trajectory with $\mathrm{Perf}(x_T) = \mathrm{Tr}(G\, M_{\text{perf}})$; and $\mathrm{Init}(x_0) = \mathrm{Tr}(G\, M_{\text{init}})$

From the Gram matrix $G^\star$ to a counter-example

Example: one step of GD on $\mathcal{F}_{0,L}$, metric $f(x_1) - f_\star \leq \tau \|x_0 - x_\star\|^2$.

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In PEPit: call `problem.solve()` then access `.eval()` to obtain values and `f.get_interpolator()` to retrieve the worst-case function values automatically.

Dimension of the counter-example

The Gram matrix $G^\star \in \mathbb{R}^{n \times n}$ (where $n =$ number of vectors, e.g. $n = 4$ for one GD step with function values).

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$\rightsquigarrow$ **Rank minimisation heuristic** (`dimension_reduction_heuristic="trace"` in PEPit):

1. First solve the SDP to get $\tau^\star$.
2. Fix the objective, add constraint $\mathrm{Tr}(G\, M_{\text{perf}}) = \tau^\star$.
3. Minimise $\mathrm{Tr}(G)$ (nuclear norm proxy for rank) subject to all constraints.
4. $\Rightarrow$ low-rank $G^\star$ $\Rightarrow$ **low-dimensional counter-example**.

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Practical outcome

Simple low-dimensional functions often result in the worst case !

— Lab 1 —

Recovering counter example functions

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Two-phase SDP:

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pepit_tau = problem.solve(
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    tol_dimension_reduction=1e-6)
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Example 1 – GD for Smooth Convex Minimization

Setting: $f \in \mathcal{F}_{0,L}$, $\quad x_{t+1} = x_t - \gamma \nabla f(x_t)$, $\quad \gamma = 1/L$, $n = 4$. Init: $\|x_0 - x_\star\|^2 \leq 1$. Perf: $f(x_n) - f_\star$.

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f = problem.declare_function(
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xs = f.stationary_point()
gs, fs = f.oracle(xs)
x0 = problem.set_initial_point()
gx, fx = f.oracle(x0)

problem.set_initial_condition((x0-xs)**2 <= 1)
for i in range(n):
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Extract:

```
xs.eval()[0]
f.get_interpolator()
```

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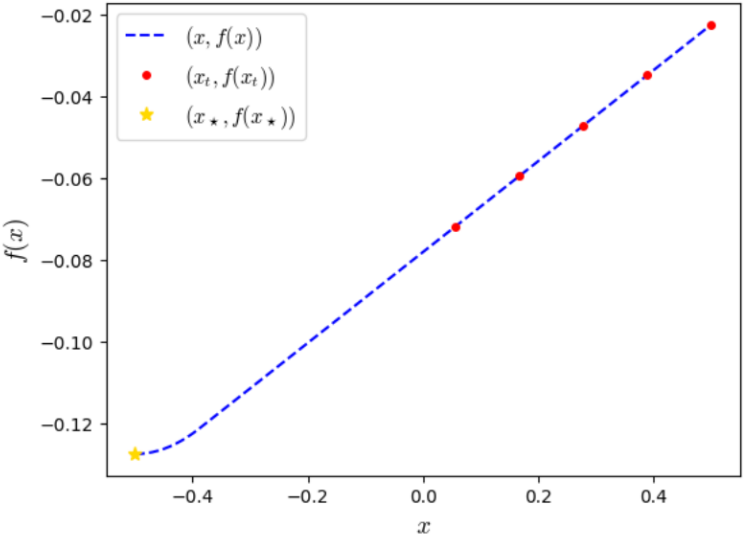
```
(PEPit) Setting up the problem: performance measure is the minimum of 1 element(s)
(PEPit) Setting up the problem: Adding initial conditions and general constraints ...
(PEPit) Setting up the problem: initial conditions and general constraints (1 constraint(s) added)
(PEPit) Setting up the problem: interpolation conditions for 1 function(s)
                        Function 1 : Adding 30 scalar constraint(s) ...
                        Function 1 : 30 scalar constraint(s) added
(PEPit) Setting up the problem: additional constraints for 0 function(s)
(PEPit) Setting up the problem: size of the Gram matrix: 7x7
(PEPit) Compiling SDP
(PEPit) Calling SDP solver
(PEPit) Solver status: optimal (wrapper:cvxpy, solver: SCS); optimal value: 0.05555881831011391
(PEPit) Postprocessing: 1 eigenvalue(s) > 1.6617368718060333e-06 before dimension reduction
(PEPit) Calling SDP solver
/usr/local/lib/python3.12/dist-packages/cvxpy/problems/problem.py:1510: UserWarning: Solution may be inaccurate. Try another solver, adjusting the solver
  warnings.warn(
(PEPit) Solver status: optimal_inaccurate (solver: SCS); objective value: 0.05555739775294698
(PEPit) Postprocessing: 1 eigenvalue(s) > 3.496552790817257 3e-11 after 1 dimension reduction step(s)
(PEPit) Solver status: optimal_inaccurate (solver: SCS); objective value: 0.055557397696808924
(PEPit) Postprocessing: 1 eigenvalue(s) > 3.516410177191783e-11 after 2 dimension reduction step(s)
(PEPit) Solver status: optimal_inaccurate (solver: SCS); objective value: 0.055557397696808924
(PEPit) Postprocessing: 1 eigenvalue(s) > 3.516410177191783e-11 after dimension reduction
(PEPit) Postprocessing: solver's output is not entirely feasible (smallest eigenvalue of the Gram matrix is: -6.69e-07 < 0).
 Small deviation from 0 may simply be due to numerical error. Big ones should be deeply investigated.
 In any case, from now the provided values of parameters are based on the projection of the Gram matrix onto the cone of symmetric semi-definite matrix.
(PEPit) Primal feasibility check:
                The solver found a Gram matrix that is positive semi-definite up to an error of 6.693242677961387e-07
                All the primal scalar constraints are verified up to an error of 1.2427625796854191e-06
(PEPit) Dual feasibility check:
                The solver found a residual matrix that is positive semi-definite up to an error of 7.128579897617394e-17
                All the dual scalar values associated with inequality constraints are nonnegative
(PEPit) The worst-case guarantee proof is perfectly reconstituted up to an error of 3.4250635130243096e-06
(PEPit) Final upper bound (dual): 0.055555127602021075 and lower bound (primal example): 0.055557397696808924
(PEPit) Duality gap: absolute: -2.270094787849841e-06 and relative: -4.0860351311599134e-05
```

Figure: Pepit output

Example 1 – GD for Smooth Convex Minimization



Example 2 – GD for Smooth (Non-Convex) Minimization

Setting: $f \in \mathcal{F}_{-L,L}$ (*no* convexity), $x_{t+1} = x_t - \gamma \nabla f(x_t)$, $\gamma = 0.95/L$, $n = 3$. Init: $f(x_0) - f_\star = 1$.
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from PEPit.functions import SmoothFunction

L = 1;  n = 3;  gamma = .95/L
problem = PEP()
f = problem.declare_function(SmoothFunction, L=L)

xs = f.stationary_point()
gs, fs = f.oracle(xs)
x0 = problem.set_initial_point()
gx, fx = f.oracle(x0)

problem.set_initial_condition(f0 - fs == 1)
for i in range(n):
    problem.set_performance_metric(gx**2)
    f.add_constraint(fs <= fx - 1/(2*L) * gx**2)
    x0 = x0 - gamma * gx
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rank-1 $\Rightarrow$ 1D

Two interpolators:
lowest / *highest* f

Example 3 – GD with Exact Line Search on $\mathcal{F}_{\mu,L}$

Setting: $f \in \mathcal{F}_{\mu,L}$, $\quad L = 1,\ \mu = 0.1, \quad \gamma_k = \arg\min_{\gamma \geq 0} f(x_k - \gamma \nabla f(x_k)), \quad n = 5.$ Init: $f(x_0) - f_\star \leq 1.$
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theoretical = ((L - mu)/(L + mu))**(2*n)
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- $\tau^*_{\text{ls}} \approx 0.1344$
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$\rightarrow$ **Line search strictly better**

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Gram matrix:

- **rank-2** $\Rightarrow$ **2D** worst-case
- Iterates *zigzag* in $\mathbb{R}^2$
- Each step $\perp$ to previous gradient

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Extract 2D:

```
x.eval()[0:2]
f.get_interpolator()
```

Take-Away: PEPit as a Counter-Example Generator

Example 1

GD, $\mathcal{F}_{0,L}$

Perf: $f(x_n) - f_\star$

rank-1 $\rightarrow$ 1D

Huber-loss function

Example 2

GD, $\mathcal{F}_{0,L}$ non-convex

Perf: $\min_t \|\nabla f(x_t)\|^2$

rank-1 $\rightarrow$ 1D

non-convex, 1D profile

Example 3

GD + line search, $\mathcal{F}_{\mu,L}$

Perf: $f(x_n) - f_\star$

rank-2 $\rightarrow$ 2D

zigzag trajectory

Take-Away: PEPit as a Counter-Example Generator

Example 1	Example 2	Example 3
GD, $\mathcal{F}_{0,L}$	GD, $\mathcal{F}_{0,L}$ non-convex	GD + line search, $\mathcal{F}_{\mu,L}$
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Huber-loss function	non-convex, 1D profile	zigzag trajectory

Three lines of interest:

1. `problem.solve(dimension_reduction_heuristic=...)` $\to \tau^*$ + low-rank G^*.
2. `point.eval()` $\to$ worst-case coordinates.
3. `f.get_interpolator()` $\to$ reconstruct worst-case function on a grid.

Take-Away: PEPit as a Counter-Example Generator

Example 1	Example 2	Example 3
GD, $\mathcal{F}_{0,L}$	GD, $\mathcal{F}_{0,L}$ non-convex	GD + line search, $\mathcal{F}_{\mu,L}$
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rank-1 $\rightarrow$ 1D	rank-1 $\rightarrow$ 1D	rank-2 $\rightarrow$ 2D
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$\rightarrow$ PEPit is not just a rate computer, it can produce *interpretable, verifiable* worst-case instances in low dimension.

References I

Appendix